The Appraisal Mechanism

Spillover Effects of All-Cash Sales on Local Housing Markets

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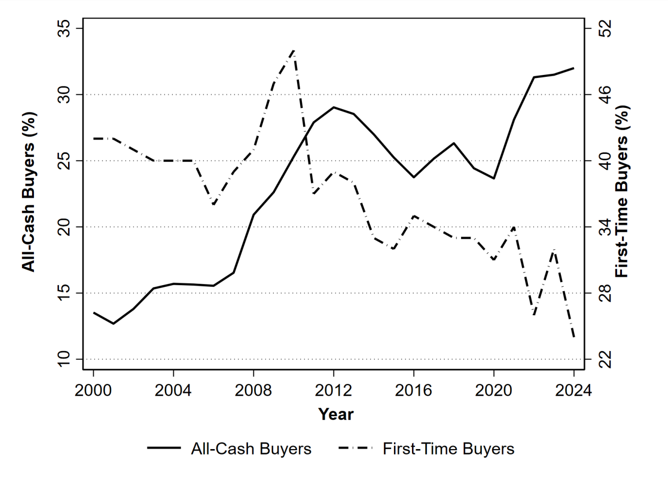
The Rise of Cash Buyers



Roughly **one in three homes** bought with all cash, while first-time buyers at an all time low of $\sim$24%

The Appraisal Mechanism

Mortgage-financed buyers pay **an 11% premium** over cash buyers
(Reher and Valcanov, JF 2024; Han and Hong, RF 2024)

Consider a mortgage-financed home **surrounded by cash sales**

1. Home appraisals rely on recent comparable sales
2. Discounted cash sales anchor appraisals **downwards** $\Rightarrow$ **less lending**
3. Buyers put more money down or walk away
4. Sellers **lower the ask price** with a depressed sale price

Institutional details

Research Question

Do nearby cash sales set lower reference prices that can spill over and constrain subsequent mortgage-financed transactions?

1. Appraisal values and transaction prices
   - Why don't buyers already anticipate this friction?
   - How much can be explained by, alternatively, seller-agent anchoring?
2. How does the effect vary by
   - Buyer income, leverage, race, loan types
   - Neighborhoods (income, race, growth)
3. Welfare implications?

Preview

1. Adopt a ring-based identification strategy (Bayer et al., AER 2021)
2. Estimate appraisal and price impact
3. A back-of-the-envelop welfare calculation

Preview

1. Adopt a ring-based identification strategy (Bayer et al., AER 2021)
   - Examine the effects of very local cash buyer activity (0.6 miles)
   - Control for comparable activity on other nearby blocks (1.2 miles)
   - **Simulate comparable sales** and validate the choice of ring

2. Estimate appraisal and price impact

3. A back-of-the-envelop welfare calculation

Preview

1. Adopt a ring-based identification strategy (Bayer et al., AER 2021)

2. Estimate appraisal and price impact
   - One SD $\uparrow$ in nearby cash sales (25) $\implies$ $\sim$1.4% $\downarrow$ in appraisal/price
   - Effects decay across distance and increase with recency
   - Effects more pronounced for **low-income, high-leverage, minority, and first-time home buyers**

3. A back-of-the-envelop welfare calculation

Preview

1. Adopt a ring-based identification strategy (Bayer et al., AER 2021)

2. Estimate appraisal and price impact
   - Neighborhoods with low recent house price growth see bigger effects than those that experienced a high growth in past five years

3. A back-of-the-envelop welfare calculation

Preview

1. Adopt a ring-based identification strategy (Bayer et al., AER 2021)
2. Estimate appraisal and price impact
3. A back-of-the-envelop welfare calculation
    - **Exclusion vs. lower cost of ownership**
    - A net welfare loss for neighborhoods with more **constrained buyers**

Roadmap

1. Contribution & literature
2. Data
3. Research design & internal validity
4. Baseline results
5. Heterogeneity
6. Welfare implications

Contribution & Literature

I identify a unique mechanism by which cash sales affect local housing market dynamics through spillover to appraisals

- Transaction-level mortgage-cash premium and its determinants
  Reher and Valcanov (2024), Han and Hong, 2024; Chia and Ambrose, 2024
  - Cash sales **spill over** into local house prices through appraisals
- The role of financial frictions in local house price discovery
  Stein, 1995; Genesove and Mayer, 2001; Landvoigt et al.; 2015; Guren, 2018
  - I investigate an **appraisal-induced** constraint of home buyers
- Dynamics and biases in appraisals and assessments
  Agarwal et al. (2017), Fout et al. (2022), Avenancio-Leon and Howard (2022)
  - I show appraisal dynamics can cause **frictions in housing evaluation**

Data

Sources: Detailed deed, tax, and listing records from CoreLogic merged with mortgage originations from HMDA **at the transaction level**

Primary Sample: 2018–2022

- Arms-length transactions; SF + townhomes
- No foreclosures or intrafamily transfers; no records with extremely low or high prices, building size, etc.
- Selection very close to Reher and Valcanov (2024)

Overview

- 6.2+ million records with transaction, listing, and loan information
- 2,074 counties and 76k tracts (90+% population)

Summary Statistics

Identifying Spillover from Nearby Cash Sales

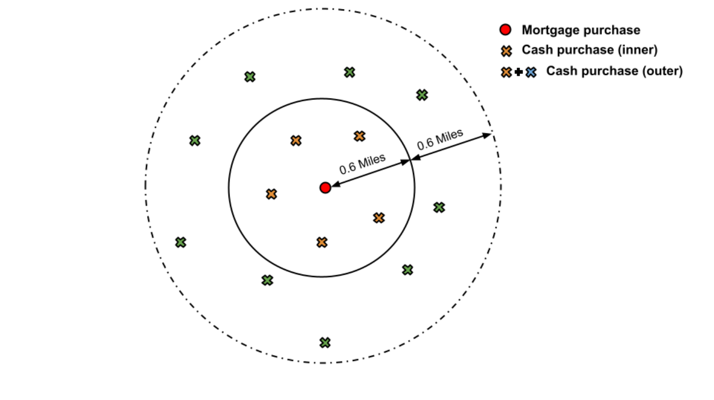
Goal: Identify causal effects of nearby cash sales on the focal home's price through depressed appraisals

Challenges:

1. Cash buyers are not randomly assigned to neighborhoods or transactions
2. Unobserved neighborhood-level factors simultaneously influence both cash buyer activity and housing outcomes

Strategy: a ring-based research design used extensively in identifying neighborhood effects and local spillovers

Ring-Based Research Design



Measure the effects of cash activity within the inner ring, while conditioning on activity in the wider band

Ring-Based Research Design

$$log(Y_{i,t}) \; = \; \beta_1 \, \mathrm{C}^{\text{inner}}_{t-s:t} + \; \beta_2 \, \mathrm{C}^{\text{outer}}_{t-s:t} + \gamma X_i \; + \; \delta_{c(i),t} \; + \; \varepsilon_{i,t}$$

- $Y_{i,t}$: appraisal or price for property i on date t
- $\mathrm{C}^{\text{inner/outer}}_{t-s:t} \;=$ **cumulative count in cash sales** within 0.6/1.2 miles in a recent time period, $t - s : t$ (s = 11 months)
- $X_{i,t}$: property, buyer, and other transaction-level characteristics
- $\delta_{c(i),t}$: tract-by-year fixed effects

β_1 is the net spillover effect on a property of having cash sales in its immediate vicinity, beyond the area-wide trends captured by β_2

Identification Assumptions

A1: The sorting of nearby cash sales is quasi-random hyper-locally

A2: Neighborhood interactions are stronger at very local geographies

Identification Assumptions

A1: The sorting of nearby cash sales is quasi-random hyper-locally

- E.g., limited ability to cherry-pick micro-locations $\implies$ a property's immediate vicinity experiencing a cash sale is largely a matter of timing and listing availability rather than a reflection of some unobserved "desirability" of that exact block
- **Testable:** The selection of cash sales into properties/blocks does not vary significantly across the geographic scale, following Bayer et al. (AER 2021)

A2: Neighborhood interactions are stronger at very local geographies

Identification Assumptions

A1: The sorting of nearby cash sales is quasi-random hyper-locally

A2: Neighborhood interactions are stronger at very local geographies

- Will find effects only if the response of focal property to cash activity is stronger within the closer vicinity than in the broader area (with implications for choosing the inner ring radius)
- Comps are predominantly drawn from the immediate vicinity so that appraisers have little reason to go outside the inner ring to find comparables

A1: Cash Sorting Does Not Vary by Geographic Scale

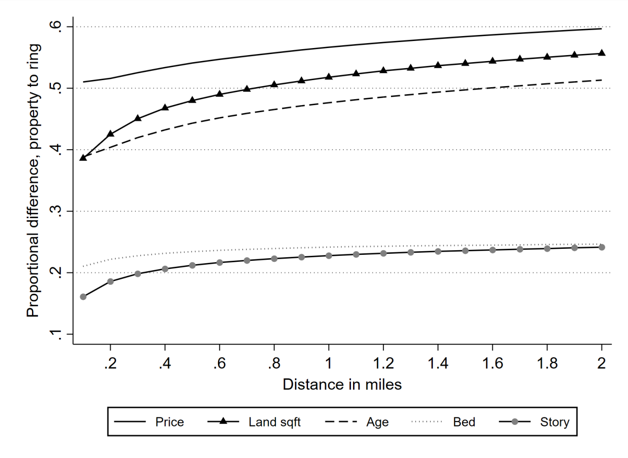
Step #1: Identify property-level selection of cash sales

- Cash buyers prefer cheaper, younger homes with fewer bedrooms, larger living space, more land and parking (conditional on tract-level characteristics) Property Selection

Step #2: Test to what degree cash sales sort by geographic proximity

- Compare a cash-purchased home's attribute x_i with the mean of those attributes within successive annuli of width d, $\bar{x}_{i,rd,(r+1)d}$ (d = 0.1 mile; r = 1, 2, ..., 20)

Selection Does Not Vary Significantly by Geographic Scale



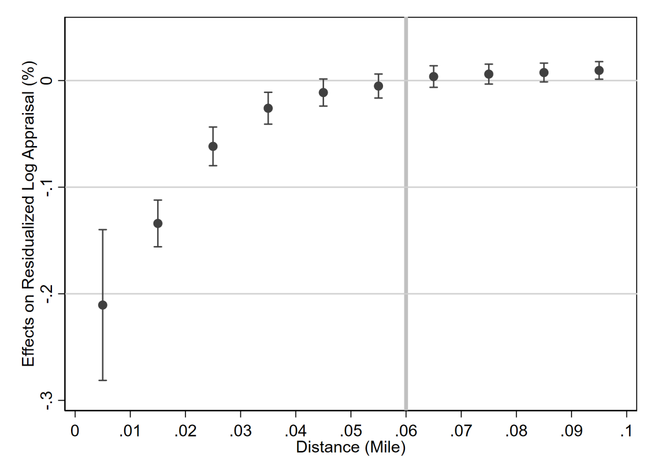
Cash-purchased properties are **only slightly less similar to their neighbors** 0.1-0.2 miles away relative to neighbors within 0.1 miles, and again slightly less to those 0.2-0.3 miles away, and so on

A2: Strong Hyper-Local Neighborhood Interactions

Test #1: Estimate the average treatment effect of nearby cash sales **in each concentric ring** (i.e., 0-0.1, 0.1-0.2, ..., 0.9-1 miles)

- See whether the effects decay in distance
- The point where the effects decay to zero suggests the choice of the inner ring radius

Decaying Average Treatment Effects



The effects decay dramatically along geographic scale and **fade around 0.6 mile** (i.e., a desirable cutoff to capture the full treatment effects)

A2: Strong Hyper-Local Neighborhood Interactions

Test #2: Are comps drawn from the more immediate vicinity?

- Limitation: no publicly available on comparable sales
- Following the industry standard, **manually construct comps** for each transaction
- Plot the number of simulated comps across distance

An Algorithm for Simulated Comparable Sales

The industry standard (e.g., Zillow) of choosing comps is based on **distance, recency, and property attributes**

- Select $\geq$3 transactions within 0.25–0.5 mile (up to 1 mile) in the past 3–6 months (up to 1 year) with similar characteristics

An Algorithm for Simulated Comparable Sales

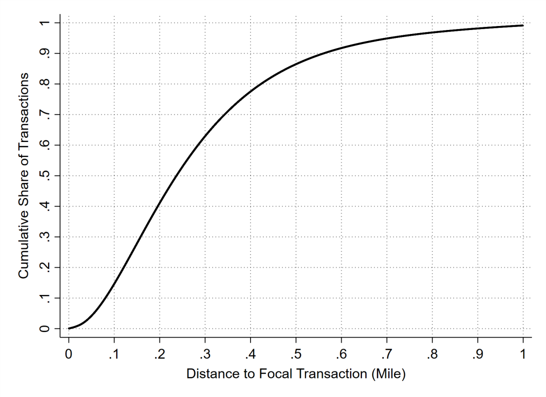
The industry standard (e.g., Zillow) of choosing comps is based on **distance, recency, and property attributes**

- Select $\geq$3 transactions within 0.25–0.5 mile (up to 1 mile) in the past 3–6 months (up to 1 year) with similar characteristics

An algorithm to manually construct comps:

1. Narrow down to potential comps traded within 1 mile & 1 year
2. Compute (dis-)similarity scores based on property attributes
   - $$S(i,j) = \sum_{k=1}^{K} w_k \cdot \frac{|x_{k,i} - x_{k,j}|}{\Delta_k}$$
   - $S(i,j)$: how dissimilar property j is to the subject property i
3. Selecting 3-4 final comps with top rankings Comps vs. Non-Comps
   - Prioritize closer, more recent candidates in the event of very close scores and similar key attributes (e.g., # bed, # stories must match)

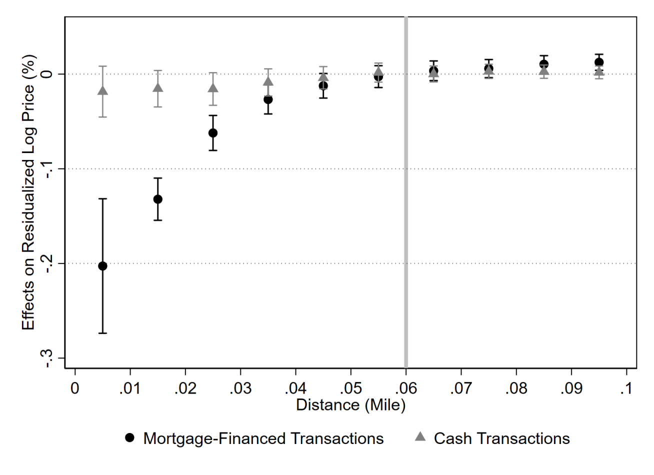
Most Imputed Comps Are Within 0.6 Mile



- **92.8%** imputed comps are within the 0.6-mile radius PDF of Imputed Comps
  - Most focal sales are matched with more than 10 nearby candidates with 1 mile → prioritizing closer and more recent candidates

Placebo Test Using Cash-Purchased Focal Properties

The impact of nearby cash sales on the prices of **a focal all-cash transaction** is negligible compared to their effects on a mortgage-financed transaction



Summary: Evidence Supporting Internal Validiity

1. **Quasi-random exposure to cash sales**
2. **Average treatment effects decay in distance**
3. **Most simulated comps are located within 0.6 mile**
4. **Placebo test shows no effects**

Summary: Evidence Supporting Internal Validiity

1. **Quasi-random exposure to cash sales**
   - Though cash buyers select at the property level, they don't significantly sort across the geographic scale
2. **Average treatment effects decay in distance**
   - Effects decay to zero around 0.6, suggesting the choice of the inner ring radius
3. **Most simulated comps are located within 0.6 mile**
   - Selected comps are indeed more similar in attributes, closer in distance and recency to the focal property
4. **Placebo test shows no effects**

Baseline Results

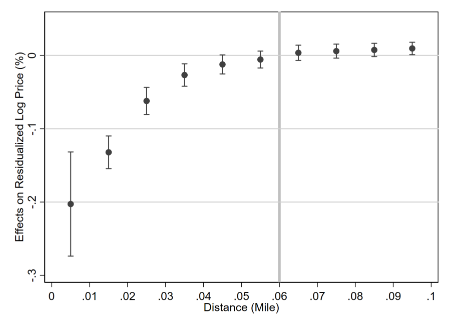
Baseline Results

	(1)	(2)	(3)	(4)	(5)	(6)
	Appraisal Values			*Transaction Prices*		
No. Cash sales						
within 0.6 miles	-0.0619***	-0.0714***	-0.0558***	-0.0619***	-0.0710***	-0.0555***
	(0.0054)	(0.0056)	(0.0047)	(0.0054)	(0.0057)	(0.0048)
within 1.2 miles	0.0002***	0.0002***	0.0001***	0.0002***	0.0002***	0.0001***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
No. Mortgage sales						
within 0.6 miles		0.0001**	0.0001**		0.0001*	0.0001*
		(0.0000)	(0.0000)		(0.0000)	(0.0000)
within 1.2 miles		-0.0000*	-0.0000*		-0.0000	-0.0000
		(0.0000)	(0.0000)		(0.0000)	(0.0000)
List price			0.5917***			0.5900***
			(0.0016)			(0.0016)
Observations	3,532,462	3,532,462	3,532,462	3,532,462	3,532,462	3,532,462
Tract-by-year FE	Y	Y	Y	Y	Y	Y
Property/Buyer/Loan Attributes	Y	Y	Y	Y	Y	Y

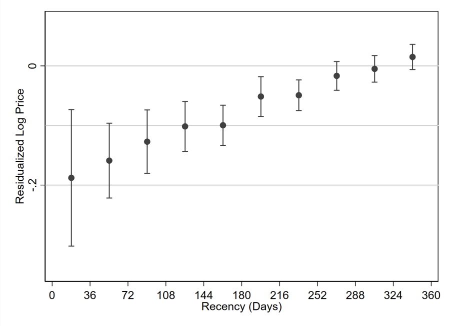
One SD $\uparrow$ in nearby cash sales (25) $\implies$ $\sim$1.4% $\downarrow$ in appraisal/price Full Table

- Nearby mortgage sales don't have any effects
- Listing price absorbs away $\sim$22% of the magnitude

Decaying Effects in Distance and Recency



A: Distance



B: Recency

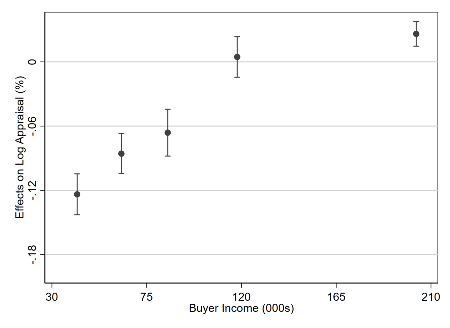
Average treatment effects decay in distance and by recency

Discussion

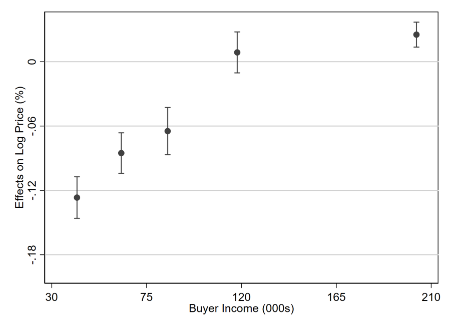
- Why don't buyers already take into account the appraisal friction when bidding?
    1. Inattention or naïveté
    2. Vulnerability to low appraisals due to inexperience, tight financial flexibility (e.g., large down-payment, low-income, tightly-budgeted, or first-time home buyers)
    3. The sequential nature of appraisals lets buyers postpone dealing with low appraisals (e.g., Calem et al. REE 2021; NAR Report, 2021)

Heterogeneity

Heterogeneity by Buyer Income



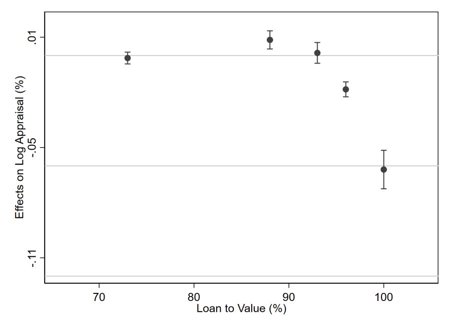
(a) Appraisal



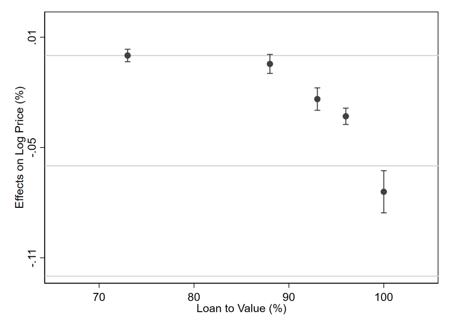
(b) Price

Effects mostly come from buyers with income $\leq$ \$100,000

Heterogeneity by Buyer Leverage



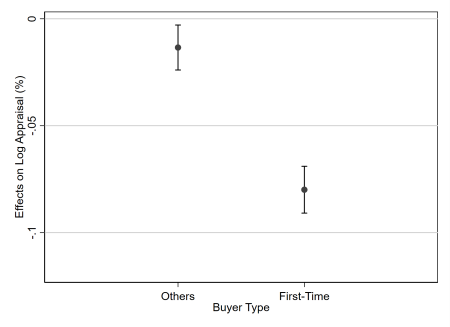
(a) Appraisal



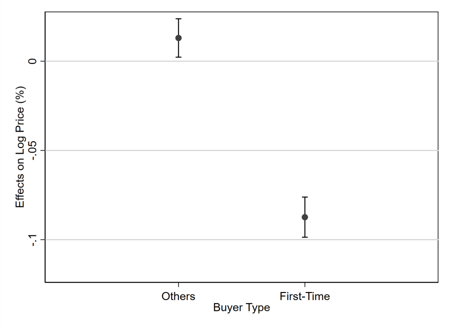
(b) Price

Effects mostly come from buyers with a loan-to-value ratio $\geq 90\%$

Effects on First-Time Buyers



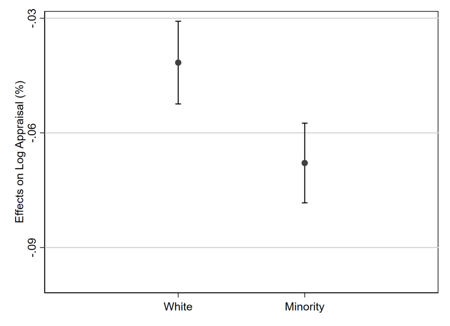
(a) Appraisal



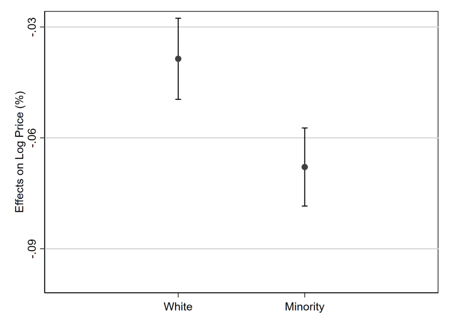
(b) Price

Larger effects on first-time buyers, compared to others with a conventional/VA/FSA/RHS loan

Effects on Minority Buyers



(a) Appraisal

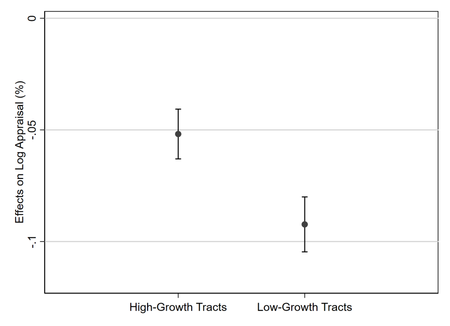


(b) Price

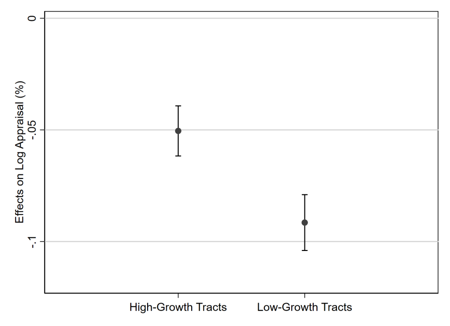
Larger effects on minority buyers [1]

[1] HMDA race categories ("derived_race"): American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian or Other Pacific Islander, White, 2 or more minority races.

Effects on High-Growth vs. Low-Growth Neighborhoods



(a) Appraisal



(b) Price

Smaller effects in areas with higher price growth[2] during 2018-2022

[2] The tract-by-year fixed effects calculated using a standard hedonic regression HPI Growth

Welfare Implications

A Back-of-the-Envelop Welfare Calculation

Lower cost of ownership vs. exclusion of constrained buyers

- Consider a neighborhood: 100 identical homes, 100 buyers
- **Two buyer types: Unconstrained vs Constrained**
- Compare two scenarios with different buyer mixes:
  - Scenario 1: 80% cash buyers, 20% constrained buyers
  - Scenario 2: 20% cash buyers, 80% constrained buyers
- Rationale:
  - Unconstrained buyers can pay without financing and often secure lower prices and take advantage of the cash spillover
  - Constrained buyers depend on mortgages and are more likely to face the appraisal constraint

Scenario 1: 80% Unconstrained / 20% Constrained

- **Unconstrained buyers:** Each pays about 10% less than financed price (\$270k vs \$300k), saving \$30k per house
  - Total surplus: $80 \times \$30\text{k} = \$2.4$ million.
- **Constrained buyers:**
  - Assume **15** buyers still manage to buy at \$300k. If each buyer values the home at \$320k, they gain \$20k surplus each (total \$300k gain)
  - Assume **5** buyers are unable to purchase and lose out on an estimated \$20k surplus each (total \$100k lost)
- **Net welfare:** \$2.6 million net benefit for all buyers
- **Outcome:** Predominantly cash buyers → lower prices and easy purchases. Most buyers benefit, with only a few constrained buyers miss out

Scenario 2: 20% Unconstrained / 80% Constrained

- **Unconstrained buyers:** Each secures a 10% discount (pays \$270k vs \$300k), saving \$30*k* per house
  - Total surplus: $20 \times \$30\text{k} = \$600\text{k}$
- **Constrained buyers:**
  - Assume **40** financed buyers successfully purchase. If each earns \$10*k* surplus, total gain $= 40 \times \$10\text{k} = \$400\text{k}$
  - Assume **40** buyers cannot buy at all and lose an estimated \$15*k* of would-be surplus each. Total lost $= 40 \times \$15\text{k} = \$600\text{k}$
- **Net welfare: \$400k** net benefit
- **Outcome:** Small net gain could turn negative if fewer constrained buyers succeed

Implications

Neighborhood composition matters:

- Different mixes of unconstrained vs constrained buyers lead to differential welfare outcomes

Welfare trade-off:

- Unconstrained buyers benefit from discounts; constrained buyers bear exclusion/appraisal-gap costs. When constrained buyers dominate, spillover costs can outweigh benefits

Implications:

- In markets with many constrained buyers, assistance aimed at appraisal gaps/down payments or improved appraisal calibration could raise overall welfare by reducing exclusions

Differential Welfare for Low- Vs. High-Growth Neighborhoods

	Low Growth / Low Demand	**High Growth / High Demand**
Net welfare	Lower: misallocation & excluded buyers	Higher: near-efficient matching
Exclusion risk	High: appraisal caps bind	Low: appraisals track market
Avg. surplus / buyer	Unequal: cash gains; constrained $= 0$ if excluded	Competitive: small per-winner surplus but full participation
Takeaway	Cash spillovers can trigger an *appraisal trap* $\Rightarrow$ welfare loss	Cash in hot markets *not* welfare-detrimental; can aid price discovery

Full Model

- **Policy lens:** In low-demand areas, reduce appraisal/down-payment frictions; in hot markets, maintain appraisal accuracy and access for first-time buyers

Conclusion

Nearby cash sales set lower reference prices that spill over and constrain subsequent mortgage-financed transactions

- Tighter financial flexibility and inexperience might be the main reason (stronger effects on more constrained home buyers)
- Weaker effects on low-growth, low-demand neighborhoods

Neighborhood composition matters for welfare: lower cost of ownership vs. exclusion

- Neighborhoods with mostly unconstrained buyers: a positive gain
- Those with many constrained buyers see a much lower gain and can even lose welfare on net

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Thank You!

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Appendix

Institutional Details on Home Appraisals

Stylized facts on home appraisals

1. In mortgage approval, lenders determine **loan amount** based on the appraisal report
2. Residential appraisals mainly rely on recent **comparable sales**
3. By regulations, the source or type of financing **must not** influence an appraisal's outcome

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Primary Sample

Table 1: Primary Sample Summary Statistics (2018–2022)

Variable	Mean	SD	Min	P25	P50	P75	Max
Sale Amount ($)	307,654	167,512	6,351	187,000	269,900	386,765	1,300,000
Appraisal Values ($)	308,224	166,754	5,000	185,000	265,000	385,000	1,005,000
Age	33	26	0	14	31	47	122
No. Bed	3.28	1	1	3	3	3	6
No. Bath	2.30	0	1	2	2	2	5
No. Stories	1.45	0	1	1	1	2	3
Land (Sqft)	16,776	21,350	1,065	6,599	9,749	16,553	168,577
Building (Sqft)	2,377	812	825	1,877	2,377	2,592	5,773
Parking (Sqft)	481	120	193	440	481	491	1281
Basement (Sqft)	750	120	120	750	750	750	1926
Income (000s)	99	61	23	57	83	124	409
LTV (%)	85	12	37	80	92	97	102
No. Observations			6,216,851				

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Property Selection of Cash Buyers

- Column (1): mortgage-cash premium ≈ 11.3%
- Column (2): property attributes predictive of cash purchases
  - Cheaper, younger homes with fewer bedrooms, larger living space, more land and parking (conditional on tract-level characteristics)

	(1) Log(Price)	(2) Cash Indicator
Cash Indicator	-0.113***	
	(0.001)	
Log(Price) Std		-0.124***
		(0.001)
Age Std	-0.091***	-0.006***
	(0.001)	(0.000)
Bed Std	0.017***	-0.005***
	(0.000)	(0.000)
Building Sqft Std	0.188***	0.038***
	(0.001)	(0.000)
Land Sqft Std	0.032***	0.009***
	(0.000)	(0.000)
Stories Std	-0.001	-0.015***
	(0.000)	(0.000)
Parking Sqft Std	0.030***	0.008***
	(0.000)	(0.000)
Basemen Sqft Std	-0.002***	-0.003***
	(0.000)	(0.000)
Observations	8,303,958	8,303,958
Tract-by-Year FE	Y	Y
Other Hedonic Controls	Y	Y
R-squared	0.795	0.161

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Summary Statistics for Each Ring

Table 2: Exposure to Nearby Cash Sales

	Panel A: Number of Cash Sales		Panel B: Number of Housing Transactions	
Distance (miles)	**Mean**	**SD**	**Mean**	**SD**
0.1	2	4	6	7
0.2	4	7	15	21
0.3	7	11	26	34
0.4	10	15	39	46
0.5	14	20	54	60
0.6	17	25	73	77
0.7	22	31	93	96
0.8	26	38	115	118
0.9	31	44	139	139
1.0	37	52	164	162
1.1	43	60	192	191
1.2	49	69	221	218
No. Observations			6,216,851	

Simulated Comps vs. Other Nearby Properties

Panel A: Summary Counts

No. Unique Pairwise Combinations	609,622,168
No. Unique Focal Transactions	3,816,516

Panel B: No. Nearby Transactions Matched Per Focal Transaction

	Mean	Std. Dev.	Min.	Max.	N
Imputed Comps	3.61	0.73	1	6	3,816,516
Other Nearby	156.11	123.57	1	2,373	3,816,516

Panel C: The Difference from Focal Transaction

	Mean	Std. Dev.	Min.	Max.	N
Group 1: Imputed Comps					
Similarity Score	0.38	0.35	0.01	3.72	13,683,225
Distance (Mile)	0.27	0.19	0	1	13,683,225
Recency (Day)	178.17	107.09	1	365	13,683,225
Building Age	5.21	9.99	0	125	13,683,225
Land Sq. ft.	4,011	19,186	0	145,547	13,683,225
Building Sq. ft.	348	396	0	2,390	13,683,225
No. Bed	0.19	1.26	0	5	13,683,225
No. Bath	0.18	0.50	0	4	13,683,225
Group 2: Other Nearby Transactions					
Similarity Score	1.19	0.50	0.01	3.72	595,938,943
Distance (Mile)	0.73	0.24	0	1	595,938,943
Recency (Day)	183.05	106.16	1	365	595,938,943
Building Age	15.24	19.09	0	125	595,938,943
Land Sq. ft.	7,583	29,128	0	189,150	595,938,943
Building Sq. ft.	831	827	0	4,727	595,938,943
No. Bed	0.72	1.90	0	5	595,938,943
No. Bath	0.73	0.94	0	4	595,938,943

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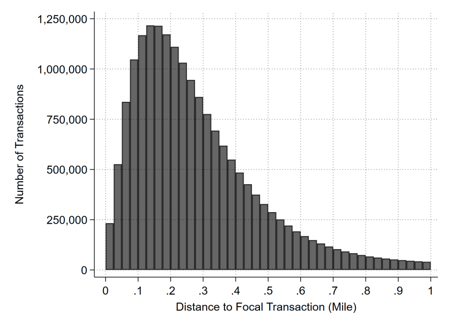
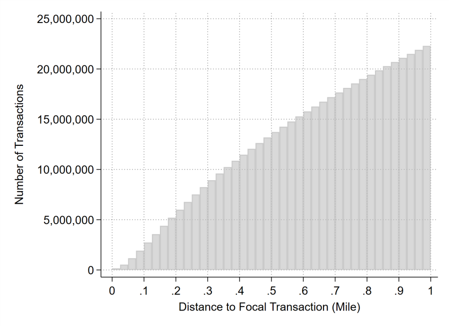
CDF of Imputed Comparables Sales



A: Histogram of Imputed Comps



B: Histogram of Other Nearby Transactions

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Baseline Results

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Appraisal Values				*Transaction Prices*			
Regression coefficients (%)								
No. Cash sales								
within 0.6 miles	-0.1356***	-0.0619***	-0.0714***	-0.0558***	-0.1355***	-0.0619***	-0.0710***	-0.0555***
	(0.0085)	(0.0054)	(0.0056)	(0.0047)	(0.0086)	(0.0054)	(0.0057)	(0.0048)
within 1.2 miles	0.0003***	0.0002***	0.0002***	0.0001***	0.0003***	0.0002***	0.0002***	0.0001***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	0.0003***	(0.0000)	(0.0000)	(0.0000)
No. Mortgage sales								
within 0.6 miles			0.0001**	0.0001**			0.0001*	0.0001*
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
within 1.2 miles			-0.0000*	-0.0000*			-0.0000	-0.0000
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
List price				0.5917***				0.5900***
				(0.0016)				(0.0016)
Constant	12.5793***	11.9753***	11.9741***	4.6383***	12.5786***	11.9856***	11.9836***	4.6689***
	(0.0008)	(0.0042)	(0.0042)	(0.0200)	(0.0008)	(0.0042)	(0.0043)	(0.0204)
Average treatment effects								
No. cash sales within 0.6 miles								
Increase by one SD	-3.38%	-1.55%	-1.93%	-1.39%	-3.37%	-1.55%	-1.91%	-1.38%
Increase from Q1 to Q3	-2.71%	-1.24%	-1.54%	-1.12%	-2.70%	-1.24%	-1.52%	-1.11%
Property characteristics		Y	Y	Y		Y	Y	Y
Buyer age	Y	Y	Y	Y	Y	Y	Y	Y
Buyer race	Y	Y	Y	Y	Y	Y	Y	Y
Loan type	Y	Y	Y	Y	Y	Y	Y	Y
Tract-by-year FE	Y	Y	Y	Y	Y	Y	Y	Y
R-squared	0.714	0.818	0.818	0.845	0.705	0.808	0.808	0.834
Observations	3,532,462	3,532,462	3,532,462	3,532,462	3,532,462	3,532,462	3,532,462	3,532,462

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Summary Statistics of HPI Growth (2018-2022)

Table 3:

Year	Mean	SD	P10	P25	P50	P75	P90	N
2018	0.064	0.211	-0.120	-0.009	0.061	0.135	0.248	66,466
2019	0.051	0.208	-0.126	-0.021	0.048	0.122	0.238	66,466
2020	0.087	0.203	-0.084	0.016	0.083	0.156	0.268	66,466
2021	0.158	0.195	-0.021	0.079	0.154	0.232	0.341	66,466
2022	0.128	0.192	-0.059	0.049	0.128	0.210	0.314	66,466
Average	0.092	0.069	0.042	0.065	0.089	0.116	0.147	66,466

This table summarizes the house price indices (HPIs) estimated from hedonic regressions and aggregated to the annual level. The last row shows the summary statistics of the five-year average price growth across all 66,466 tracts.

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Model Setup: Neighborhood Welfare Tradeoff

- **Environment:** Neighborhood with N houses and two buyer types:
  - **Unconstrained (cash):** No financing frictions, can pay in full
  - **Constrained (mortgage):** Face **appraisal cap** $P_i \leq W_i + \lambda A_t$
- **Cash spillover effect:** Nearby cash transactions at discounted prices $\downarrow$ appraised values A_t
- **Welfare channels:**
  - **Benefit:** Cash buyers gain surplus from lower purchase price
  - **Cost:** Constrained buyers excluded or forced to bring extra equity
- **Tradeoff:** Net welfare depends on buyer composition:
  - More cash buyers $\Rightarrow$ larger price discounts, fewer excluded
  - More constrained buyers $\Rightarrow$ more exclusion, lower aggregate welfare

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Low Growth / Low Demand Neighborhood

- **Market conditions:** Weak demand, sparse sales $\Rightarrow$ discounted cash transactions push down appraisals A_t
- **Financing binds:** Low $A_t \Rightarrow$ tight mortgage cap $P_i \leq W_i + \lambda A_t \ll v_i$
- **Tradeoff: Cheaper prices** benefit cash buyers (buy at $P < v_i$); **exclusion/misallocation** for constrained high-v buyers (cannot bridge appraisal gap).
- **Dynamic:** Appraisal constraint – discounted $P_t \downarrow \Rightarrow A_{t+1} \downarrow \Rightarrow$ future P_{t+1} capped
- **Welfare:** Lost surplus from excluded buyers + misallocation (home not going to highest-v) $\Rightarrow$ **lower neighborhood buyer welfare** Back

High Growth / High Demand Neighborhood

- **Market conditions:** Strong demand, rapid turnover; frequent high-P comps keep A_t in line with fundamentals
- **Frictions minimal:** $P_i \leq W_i + \lambda A_t \approx v_i$; buyers can bid near true v (cash or with appraisal-gap coverage)
- **Outcome:** Bidding competes prices to v; allocation close to efficient; little exclusion at the micro level
- **Welfare:** Nearly all buyer surplus realized via efficient matching; cash presence does **not** depress A_t and can speed price discovery

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